EXPLAINING IMAGE COLORIZATION: A REGION-BASED XAI STUDY OF DDCOLOR

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Project video: youtube.com/xai_project_video
Drive Link: drive.google.com/xai_project
Baseline code: github.com/piddnad/DDColor
Weights: huggingface.co/piddnad/ddcolor_paper_tiny
Datasets: COCO val2017 ADE20K validation

ABSTRACT

Image colorization can produce realistic colors from a grayscale image, but the output alone does not show which image regions affected the color decision. We study DDColor, an ICCV 2023 colorization model, using two main XAI methods: Region-Grad-CAM and Region-RISE. Query attention and semantic-region occlusion are also used as supporting analyses. We evaluate fixed 150-image subsets of COCO and ADE20K with colorization, localization, faithfulness, sanity-check, and runtime metrics. Region-Grad-CAM gives the best overall localization and insertion results. Semantic occlusion shows that the selected region is usually the main source of its color, while surrounding context is the strongest source in 15.3% of ADE20K images.

1 INTRODUCTION

Automatic image colorization estimates missing colors from a grayscale image. The task is ambiguous because the same grayscale pattern may have several reasonable colors. A model may therefore create a realistic image while relying on the wrong object, texture, or surrounding region. Our main question is: *which input regions affect the predicted color of a selected output region?*

We investigate this question using DDColor (Kang et al., 2023), a photo-realistic colorization model introduced at ICCV 2023. Unlike classification or object detection, colorization produces dense chroma values, so standard explanation targets must be adapted. DDColor's two decoders combine local image detail with scene-level information, making it suitable for studying both local and contextual color evidence.

Our two main XAI methods are Region-Grad-CAM, adapted from Grad-CAM (Selvaraju et al., 2017), and Region-RISE, adapted from RISE (Petsiuk et al., 2018). Both explain the predicted color inside a selected region rather than a classification score. We also inspect DDColor's query attention and apply semantic occlusion on ADE20K. COCO is used for general colorization analysis, while ADE20K supports region-based semantic analysis.

2 METHODS

2.1 DDCOLOR BASELINE

DDColor predicts the a and b color channels in CIELAB space from the grayscale luminance channel L. As shown in Figure 1, the pixel decoder preserves spatial details, while the color decoder uses learned color queries and attention to represent semantic color information. A fusion module combines both outputs to produce the final colorization (Kang et al., 2023). To achieve an optimal balance between computational efficiency and colorization quality based on our initial trials, we opted for the official `ddcolor_paper_tiny` checkpoint. Images are resized to 256×256 before inference.

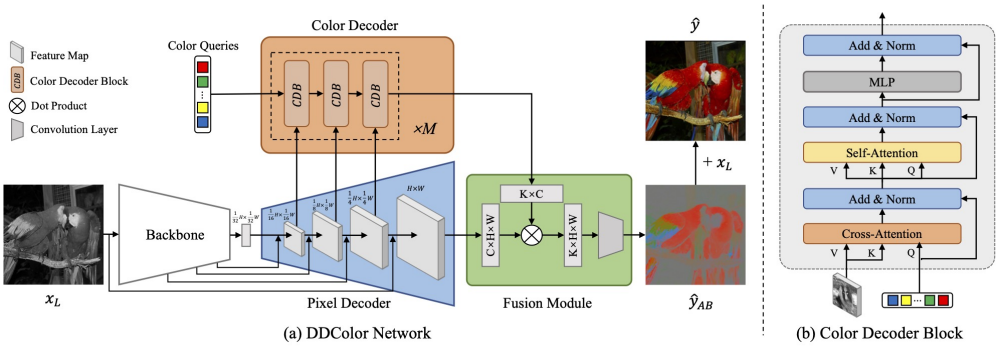


Figure 1: DDColor network and color decoder block. The figure is reproduced from the official DDColor repository and paper (Kang et al., 2023).

2.2 Region-specific explanation target

Standard Grad-CAM and RISE usually explain a classification logit. DDColor instead outputs a dense tensor $\hat{\mathbf{y}} \in \mathbb{R}^{2\times H\times W}$. For a binary region mask M, we define the scalar target as mean predicted chroma:

$$S_M(\hat{\mathbf{y}}) = \frac{1}{|M|} \sum_{(i,j)\in M} \sqrt{\hat{a}_{ij}^2 + \hat{b}_{ij}^2 + \epsilon}. \tag{1}$$

This score represents the average predicted color strength inside the selected mask. It allows Grad-CAM, RISE, and attention maps to explain the same regional output.

2.3 Region-Grad-CAM

Grad-CAM computes the gradient of the target with respect to activations A^k in a selected convolutional layer. In our implementation, the target layer is the last spatial convolutional block available before the final chroma prediction head. Channel weights are obtained by global average pooling,

$$\alpha_k = \frac{1}{Z} \sum_{u,v} \frac{\partial S_M}{\partial A_{uv}^k}, \tag{2}$$

and the map is

$$L_{\mathrm{GC}} = \mathrm{ReLU}\left(\sum_k \alpha_k A^k \right). \tag{3}$$

The map is resized to the input resolution and normalized. Grad-CAM is fast because it needs one forward and backward pass, but the result depends on the selected internal layer and may be spatially coarse.

2.4 Mean-centered Region-RISE

RISE is a black-box method that repeatedly masks the input and weights each mask by the resulting target score. We generate $N = 64$ binary masks on an 8×8 grid, upsample them with random shifts, and use visibility probability $p = 0.5$. Hidden pixels are replaced by the image mean $\bar{x}$. For mask m_n, the perturbed input and score are

$$\tilde{x}_n = x \odot m_n + \bar{x}(1 - m_n), \qquad s_n = S_M(f(\tilde{x}_n)). \tag{4}$$

The finite-sample centered map is

$$L_{\mathrm{RISE}} = \mathrm{ReLU}\left[\frac{1}{N} \sum_{n=1}^{N} s_n m_n - \left(\frac{1}{N} \sum_{n=1}^{N} s_n \right) \left(\frac{1}{N} \sum_{n=1}^{N} m_n \right) \right]. \tag{5}$$

The centering term removes the average effect of mask visibility. Region-RISE does not require gradients or access to internal layers, but it is slower and can be noisy. We use 64 masks to keep the full experiment practical.

2.5 Supplementary analyses: query attention and semantic occlusion

Region-conditioned query attention. DDColor contains cross-attention layers between learned color queries and image features. We combine the final three attention layers and give more weight to queries that affect the selected regional score. The resulting map summarizes where the decoder's color queries attend for the selected region. Attention is treated as supporting evidence, not as a complete explanation by itself (Jain and Wallace, 2019).

Semantic-region occlusion. For ADE20K, we hide one complete semantic region at a time by replacing it with the input mean. We then measure how much the predicted color changes inside the selected target region. A large change means that the hidden region influenced the target. Regions are grouped as sky, road/path, vegetation, built surface, and object/other.

2.6 Explanation evaluation

Localization. Energy inside the mask measures how much of the explanation falls inside the selected region. Pointing accuracy checks whether the strongest explanation pixel is inside that region.

Deletion and insertion. Deletion gradually removes the most important pixels; lower AUC is better because the prediction should change quickly. Insertion gradually restores important pixels into a mean image; higher AUC is better because the original prediction should recover quickly.

Sanity check. Following Adebayo et al. (2018), we randomize the model parameters and recompute the maps. A low correlation with the trained-model maps shows that an explanation depends on learned parameters.

3 Experimental Settings

Dataset roles. COCO val2017 and ADE20K validation have different roles. COCO tests general colorization on diverse real-world scenes containing people, animals, vehicles, food, and other objects. ADE20K supports semantic XAI analysis because it provides dense labels for regions such as sky, road, vegetation, wall, water, and buildings. For both datasets, RGB images are converted to grayscale inputs and the original RGB images are used as references.

Group responsibilities. One team member focused on the COCO val2017 experiments and Region-Grad-CAM analysis. The other team member focused on the ADE20K validation experiments, Region-RISE, and semantic occlusion analysis. Both members contributed to DDColor inference, metric computation, and the comparison of the results.

Datasets and sampling. For COCO val2017, we select the largest non-crowd polygon occupying 5–70% of image area. For ADE20K validation, we select the largest valid semantic class in the same range. A reproducible selection protocol provides 150 valid images from each dataset. Localization is evaluated on all selected images, faithfulness on 30 images per dataset, and parameter randomization on 10 images per dataset. Semantic occlusion uses all 150 ADE20K images, with up to eight source classes covering at least 2% each, producing 805 interventions.

Environment. The experiments were conducted using an NVIDIA RTX 4050 GPU and Google Colab. The input resolution for DDColor is 256×256. RISE is configured with 64 masks, a batch size of 4, an 8×8 grid, and a visibility probability of $p = 0.5$. The query attention mechanism combines the final three decoder layers.

Colorization metrics. PSNR measures pixel fidelity, and SSIM measures structural fidelity; higher is better for both. CIEDE2000 measures perceptual color difference, LPIPS measures learned perceptual distance, and absolute colorfulness error compares the predicted and reference image colorfulness; lower is better for these three metrics. Because colorization is multimodal, reference-based scores may penalize plausible colors that differ from the single ground truth. We therefore include qualitative outputs.

4 Experimental Results

4.1 Colorization performance

Table 1: DDColor quality on the fixed subsets (mean ± standard deviation). CF is absolute colorfulness error.

Dataset	PSNR ↑	SSIM ↑	ΔE_{00} ↓	LPIPS ↓	CF ↓
COCO	22.920 ± 4.179	0.899 ± 0.063	9.835 ± 3.747	0.155 ± 0.064	11.406 ± 10.502
ADE20K	24.164 ± 3.274	0.936 ± 0.038	9.388 ± 3.221	0.136 ± 0.051	10.845 ± 9.168

Table 1 shows stable DDColor performance on both subsets. ADE20K has slightly better average results, including a 1.24 dB higher PSNR and lower LPIPS. Since the two subsets contain different scenes, this comparison should not be interpreted as proof that ADE20K is always easier.

4.2 Quantitative XAI comparison

Table 2: XAI results (mean ± standard deviation). Faithfulness uses 30 images per dataset; localization uses 150.

Dataset	Method	Energy ↑	Pointing ↑	Deletion ↓	Insertion ↑
COCO	Grad-CAM	0.379 ± 0.281	0.460 ± 0.500	$\mathbf{0.398} \pm 0.149$	$\mathbf{0.475} \pm 0.166$
COCO	RISE	0.144 ± 0.150	0.173 ± 0.380	0.447 ± 0.149	0.406 ± 0.155
COCO	Query-Attn	0.221 ± 0.149	0.133 ± 0.341	0.417 ± 0.145	0.422 ± 0.140
ADE20K	Grad-CAM	$\mathbf{0.590} \pm 0.236$	$\mathbf{0.540} \pm 0.500$	0.447 ± 0.165	$\mathbf{0.530} \pm 0.135$
ADE20K	RISE	0.373 ± 0.182	0.347 ± 0.478	$\mathbf{0.410} \pm 0.182$	0.488 ± 0.151
ADE20K	Query-Attn	0.321 ± 0.143	0.240 ± 0.429	0.453 ± 0.149	0.442 ± 0.178

Region-Grad-CAM has the strongest localization. Its mean energy across both datasets is 0.485, compared with 0.259 for RISE and 0.271 for Query-Attn. Grad-CAM also has the best overall insertion score. RISE gives the best deletion score on ADE20K, but it is much slower. Query attention gives useful additional information, although its localization and faithfulness are generally weaker than Grad-CAM. Its low agreement with the other maps also shows that attention captures a different internal signal.

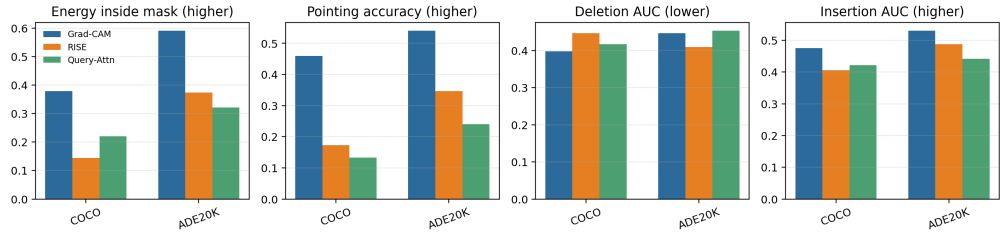


Figure 2: Comparison of gradient, perturbation, and query-attention maps. Attention is more localized than RISE on COCO but not ADE20K, and Grad-CAM has the strongest insertion performance.

4.3 Qualitative interpretation

Figure 3 contrasts the two main saliency maps with the supplementary query-attention map. High heat inside the mask indicates agreement with the evaluated region, but outside heat is not automatically erroneous because indoor materials may be inferred from nearby lamps, windows, and walls. Query attention is sparse around several contextual cues and differs visibly from the broader Grad-CAM response and the perturbation-based RISE map.

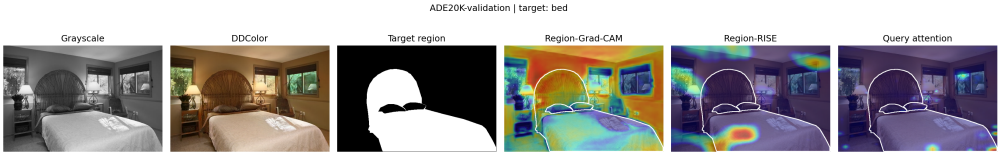


Figure 3: ADE20K bed example. Query attention visualizes decoder attention for the selected region and is used as supplementary evidence, not as a replacement for the two main XAI methods.

4.4 Semantic region influence

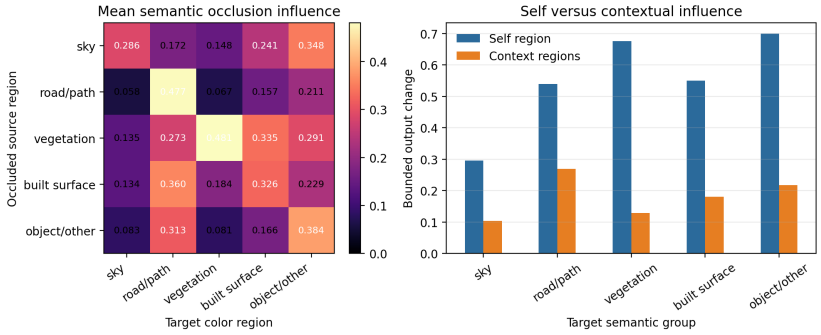
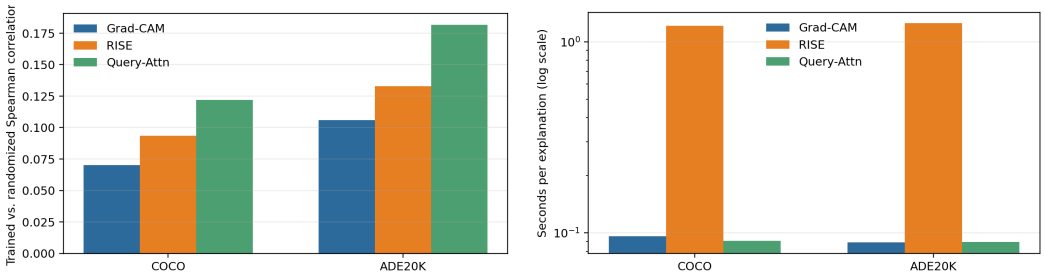


Figure 4: ADE20K semantic occlusion. Left: source-group to target-group influence, including self interventions. Right: self versus mean contextual influence for each target group.

Across 805 interventions, the mean influence of the target region on itself is 0.552, while the mean influence of surrounding regions is 0.186. A surrounding region is the strongest source in 23 of 150 images (15.3%). Examples include tree regions affecting buildings and building regions affecting roads. DDColor therefore usually uses the target region itself, but scene context can also affect its color decision.

4.5 Parameter sensitivity and efficiency



(a) Sanity-check correlation

(b) Runtime per explanation

Figure 5: All maps change after parameter randomization. Query attention is close to Grad-CAM in runtime, while RISE requires many model queries.

All trained-versus-randomized correlations are low: 0.070–0.122 on COCO and 0.106–0.182 on ADE20K. This suggests that the maps depend on learned model parameters. The sanity test uses 10 images per dataset, so it is a supporting check rather than a complete evaluation.

Grad-CAM takes about 0.093 s per explanation, Query-Attn takes 0.090 s, and RISE takes 1.23 s. Grad-CAM and attention are therefore suitable for quick inspection, while RISE is more expensive.

5 Discussion and Conclusions

The main methodological challenge was adapting classification-oriented XAI methods to a dense colorization output. We addressed this by defining a shared regional chroma target, which allows Grad-CAM, RISE, and query attention to explain the same selected output region. A second challenge was separating direct evidence from scene context. ADE20K semantic masks and region-level occlusion made it possible to measure these effects separately.

Region-Grad-CAM is the most practical main method because it gives the strongest localization and insertion results with low runtime. Region-RISE can explain the model without using its internal layers and performs well on some deletion tests, but it is slower and noisier. Query attention helps show where the color decoder looks, but it should be used together with Grad-CAM or occlusion results.

Semantic occlusion helps explain why heat outside the target mask can still be meaningful: in 15.3% of ADE20K images, another region causes the largest change. However, hiding a region with its mean value can create an unnatural input, and larger regions may have a larger effect simply because more pixels are removed.

Together, the XAI methods provide complementary information at three levels. Grad-CAM identifies the internal features that support a regional color decision, RISE checks the same decision through black-box perturbations, and attention analysis shows how information is distributed inside the color decoder. Semantic occlusion adds a scene-level view by measuring whether the target region or its surrounding context has the stronger influence. This combination helps distinguish direct target evidence from useful contextual evidence and makes the interpretation more reliable than a single heatmap.

The explanation pipeline can be strengthened by using a denser RISE mask bank, comparing multiple Grad-CAM layers to select the most informative feature level, and adding hue-specific regional targets in addition to chroma strength. Blurred or inpainted replacements can also make occlusion inputs more natural than mean-value filling. These changes would reduce perturbation noise, improve region comparisons, and provide a more detailed explanation of the model's color choices.

In conclusion, DDColor usually relies on local evidence, but nearby objects and scene context can also affect the selected region's color. Grad-CAM gives the strongest main explanation, RISE provides a black-box check, attention shows information flow inside the decoder, and semantic occlusion measures context effects. Using these methods together gives a clearer interpretation than using a single heatmap.

References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 3543–3556, 2019.

Xiaoyang Kang, Tao Yang, Wenqi Ouyang, Peiran Ren, Ling Li, and Xuansong Xie. DDColor: Towards photo-realistic image colorization via dual decoders. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 328–338, 2023.

Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. In *British Machine Vision Conference*, 2018.

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626, 2017.